Since drawing a spade and 2 knaves should occur together,

No. of ways drawing one spade and two knave cards = $n(A_2)$ = 13×78 =1014

$$\text{Therefore, } P(A_2) = \frac{n(A_2)}{n(S)} = \frac{13 \times 78}{22100}$$

$$\text{Hence, } P(A_2) = \frac{1014}{22100} = \frac{507}{11050}$$

(c) Let A_3 = drawing one spade, one knave and one heart cards

No. of ways of drawing one spade, one knave and one heart cards is

$$n(A_3) = 13C_1 \times 13C_1 \times 13C_1 = 13 \times 13 \times 13$$

$$\text{Therefore, } P(A_3) = \frac{n(A_3)}{n(S)} = \frac{13 \times 13 \times 13}{22100}$$

$$\text{Hence, } P(A_3) = \frac{2197}{22100}.$$

8.4 Addition Theorem of Probability

Theorem 8.4 : (Addition Theorem of Probability for Two Events)

If A and B are any two events in a random experiment, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Proof:

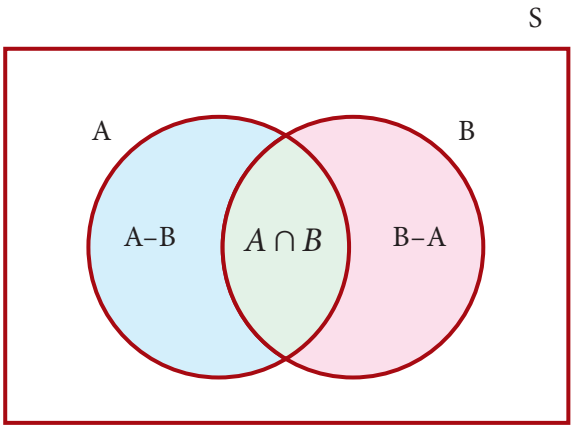


fig. 8.9 Venn diagram

For any two events A and B, the shaded region in fig. 8.9 represents the event A ∪ B.

$$A \cup B = A \cup (B - (A \cap B))$$

The events A and B-(A ∩B) are mutually exclusive.

Using Axiom 3,

$$P(A \cup B) = P(A \cup [B-(A \cap B)])$$

$$= P(A) + P[B-(A \cap B)] \qquad (8.1)$$